

ARCAS

Anti-Corruption & Accountability System

Document 2 of 4 | Version 3.0
Environment Setup Guide - Verified Operational

Author	Martin Galan
Version	3.0 - Post-implementation verified
Status	Operational - smoke test 7/7 PASS
Host	Linux Mint, 16 GB RAM, 8 CPU cores
Project Root	/home/pruebas/formacion/arcas
GitHub	github.com/mgalanme/arcas
Python	3.11 uv venv manager
Language	British English

Legal Disclaimer

Personal agentic AI learning project. Educational purposes only. Opinions personal, not those of any employer.

1. Architecture and Decisions

1.1 Three-Tier Data Architecture

Tier	Location	Purpose	Technologies
1 - Operational	Containers on laptop (Docker Compose / k3s)	Real-time ingestion, stream processing, knowledge graph, agentic detection, HITL interface	Kafka KRaft, Flink, Neo4j, PostgreSQL, Qdrant, LangGraph, Streamlit
2 - Lakehouse	MinIO on laptop (Iceberg tables)	Historical storage, CDC sink, analytical base, Databricks source	Apache Iceberg + Debezium + Flink
3 - Analytics	Databricks Community Edition (cloud)	Advanced analytics, ML training, collaborative notebooks	Databricks + Unity Catalog + DuckDB

1.2 Architectural Decision Records (ADRs)

ADR-009: Infisical Dropped - OOM on Development Laptop

Field	Content
Status	Accepted - Closed
Context	Infisical image is 2.49 GB. With full stack running (12 GB used of 16 GB), the OOM killer terminated the container (exit code 137) repeatedly.
Decision	Infisical removed from development stack. Secrets via .env (gitignored) for development. Kubernetes native Secrets for k3s workloads. Infisical remains viable for production on a server with adequate RAM.
Consequence	Simpler stack, no OOM risk. Acceptable for an educational project on a constrained laptop.

ADR-010: Solace Agent Mesh Deferred to Document 3

Field	Content
Status	Accepted - Deferred
Context	SAM requires ~1 GB RAM. With the full stack running, RAM headroom is insufficient for a stable SAM deployment.
Decision	SAM deferred until the agent layer (Document 3) is implemented. No value running a governance bus before the agents that publish to it exist. SAM integrated in Document 3 Phase 3.
Consequence	Agent governance events logged directly to PostgreSQL audit_log via audit_service.py in the interim.

ADR-011: Qdrant Healthcheck via bash /dev/tcp

Field	Content
Status	Accepted
Context	The Qdrant container image contains neither curl nor wget. Standard healthcheck commands fail with command not found.
Decision	Use bash built-in TCP redirection: exec 3<>/dev/tcp/localhost/6333 combined with grep -q passed. Zero external binary dependencies.

Field	Content
Consequence	Non-obvious healthcheck syntax. Documented here to prevent regression on image updates.

ADR-012: Redpanda Console Config Schema (v3.7.4)

Field	Content
Status	Accepted
Context	Redpanda Console v3.7.4 changed its YAML schema. kafka.schemaRegistry and connect top-level keys are no longer valid, causing crash on startup.
Decision	Updated config: schemaRegistry and kafkaConnect moved to top-level (not nested under kafka).
Consequence	Config must be verified against installed console version. Pin image version for stability.

2. Verified Operational Status

Environment Status

Smoke test passed 7/7 on 01 June 2026. All core services healthy. Ready for Document 3 implementation.

Component	Status	Details	Verified
PostgreSQL 16	healthy	14 tables, 4 triggers, 8 seed sources, audit append-only	<code>docker exec arcas-postgres psql -U arcas_app -d arcas -c '\dt'</code>
Neo4j 5.23 + APOC	healthy	8 constraints, 33 indexes, 437 APOC procedures, ontology node	APOC verified in init script
Qdrant	healthy	<code>bash /dev/tcp/healthcheck.0</code> collections (ready for use)	<code>curl localhost:6333/healthz</code>
MinIO	healthy	6 buckets, lifecycle policies, Iceberg warehouse path	<code>mc ls arcas/</code>
Redis	healthy	maxmemory 200mb, allkeys-lru policy	<code>redis-cli ping</code>
Kafka KRaft	healthy	30 topics, KRaft mode (no ZooKeeper)	<code>kafka-topics --list</code>
Apicurio Schema Registry	healthy	Schema Registry v2 API compatible	HTTP 200 on :8081
Redpanda Console	up	No healthcheck. Verified visually at localhost:8082	Browser: 1 broker online, 30 topics
Infisical	DROPPED	OOM - see ADR-009. Use .env for secrets.	N/A
Solace SAM	DEFERRED	Integrated in Document 3 - see ADR-010	N/A
LangGraph	importable	Verified via smoke test	<code>test_agent.py --synthetic</code>
Groq API	configured	Key set in .env. Live test skipped (<code>--skip-llm</code>)	<code>test_agent.py --synthetic --skip-llm</code>

2.1 RAM Profile (Verified)

Container	RAM Used	Limit	Profile
arcas-neo4j	~759 MB	1.5 GB	storage
arcas-kafka	~386 MB	1.0 GB	messaging
arcas-apicurio	~185 MB	256 MB	messaging
arcas-minio	~107 MB	512 MB	storage
arcas-redpanda-console	~27 MB	128 MB	messaging
arcas-qdrant	~17 MB	512 MB	storage
arcas-postgres	~23 MB	512 MB	storage
arcas-redis	~5 MB	256 MB	storage
arcas-registry	~11 MB	128 MB	registry
TOTAL (storage +	~1.52 GB		Well within budget

Container	RAM Used	Limit	Profile
messaging)			

3. File Inventory

3.1 Infrastructure and Configuration Files

File Path	Purpose
<code>docker-compose.yml</code>	Complete file: 7 profiles, 22 services, resource limits, healthchecks
<code>pyproject.toml</code>	pytest pythonpath, ruff linter, mypy
<code>.env.example</code>	All required environment variables. Copy to <code>.env</code>
<code>.gitignore</code>	Excludes <code>.env</code> , <code>venvs</code> , <code>data</code> , <code>model weights</code>
<code>Makefile</code>	All standard operations: <code>setup</code> , <code>start</code> , <code>stop</code> , <code>test</code> , <code>lint</code> , <code>demo</code> , <code>backup</code> , <code>push</code>
<code>README.md</code>	Project overview, quick start, architecture, legal disclaimer

File Path	Purpose
<code>config/postgres/init.sql</code>	14 tables, indexes, enum types, 8 seed sources, REVOKE permissions
<code>config/postgres/audit_trigger.sql</code>	Append-only triggers on <code>audit_log</code> , auto-update timestamps
<code>config/qdrant/config.yaml</code>	Qdrant config. Note: healthcheck uses <code>bash /dev/tcp</code> (see ADR-011)
<code>config/kafka/redpanda-console-config.yaml</code>	Updated schema for Console v3.7.4 (see ADR-012)
<code>config/flink/flink-conf.yaml</code>	Memory bounds: JobManager 512m, TaskManager 1024m
<code>config/grafana/datasources.yaml</code>	Prometheus, Loki, Tempo data sources with trace correlation
<code>config/grafana/dashboards.yaml</code>	Dashboard provisioning config
<code>config/grafana/dashboards/arcas_overview.json</code>	Placeholder dashboard
<code>config/loki/loki-config.yaml</code>	Single-node, filesystem, 15-day retention
<code>config/loki/promtail-config.yaml</code>	Docker log shipper labelled by service name
<code>config/otel/otel-collector-config.yaml</code>	OTLP receiver, memory_limiter, fanout to Prometheus/Loki/Tempo
<code>config/prometheus/prometheus.yml</code>	Scrape config for all ARCAS services
<code>config/prometheus/alertmanager.yaml</code>	Dev-log receiver (no external notifications)
<code>config/tempo/tempo.yaml</code>	OTLP receiver, local filesystem, 72h retention
<code>config/infisical/docker-compose.infisical.yml</code>	REFERENCE ONLY - not used (ADR-009). Kept for future production use.
<code>config/sam/docker-compose.sam.yml</code>	DEFERRED - integrated in Document 3 (ADR-010)

3.2 Setup Scripts

File Path	Purpose
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File Path	Purpose
scripts/setup/deploy_config_files.sh	Places all downloaded config files into correct project paths. Run once after download.
scripts/setup/pull_and_cache_images.sh	Pulls Docker images from Hub, caches in local registry. Run once before development.
scripts/setup/create_venvs.sh	Creates .venv-langchain, .venv-crewai, .venv-dashboard. spaCy>=3.8.0,<4.0 (see note below).
scripts/setup/create_kafka_topics.sh	Creates 30 Kafka topics with partitions and retention policies. Verified operational.
scripts/setup/init_minio_buckets.sh	Creates 6 MinIO buckets with GDPR-aligned lifecycle policies. Verified operational.
scripts/setup/init_neo4j_schema.py	8 constraints, 33 indexes, full-text indexes, APOC verification. Verified operational.
scripts/setup/init_sam_topics.sh	Configures Solace SAM via SEMP API. DEFERRED - run when SAM is deployed in Document 3.

spaCy Version Note

spacy==3.7.6 is yanked (incorrect transformer compatibility). create_venvs.sh uses spacy>=3.8.0,<4.0. This is verified and corrected.

3.3 Maintenance Scripts

File Path	Purpose
scripts/maintenance/backup.py	pg_dump PostgreSQL + Neo4j Cypher export to MinIO arcas-backups bucket
scripts/maintenance/test_ingestion.py	Smoke test: synthetic messages to Kafka raw topics
scripts/maintenance/test_agent.py	Smoke test: module imports, DB connections, LangGraph structure. 7/7 PASS verified.

4. Setup Sequence - Current Status

Step	Action	Script / Command	Status
1	Prerequisites installed	Manual	DONE
2	Repository cloned	git clone	DONE
3	.env configured	cp .env.example .env	DONE
4	Local registry started	docker compose --profile registry up -d	DONE
5	Docker images cached	scripts/setup/pull_and_cache_images.sh	DONE
6	Python venvs created	scripts/setup/create_venvs.sh	DONE
7	Storage profile started	docker compose --profile storage up -d	DONE - all healthy
8	PostgreSQL schema	Automatic via init.sql	DONE - 14 tables verified
9	Neo4j schema	scripts/setup/init_neo4j_schema.py	DONE - 8 constraints, 33 indexes
10	Messaging profile started	docker compose --profile storage --profile messaging up -d	DONE - all healthy
11	Kafka topics created	scripts/setup/create_kafka_topics.sh	DONE - 30 topics verified
12	MinIO buckets created	scripts/setup/init_minio_buckets.sh	DONE - 6 buckets verified
13	Smoke test passed	scripts/maintenance/test_agent.py --synthetic --skip-llm	DONE - 7/7 PASS
14	Infisical	config/infisical/docker-compose.infisical.yml	DROPPED - ADR-009
15	Solace SAM	config/sam/docker-compose.sam.yml + init_sam_topics.sh	DEFERRED - ADR-010
16	k3s namespaces	deploy/base/namespaces.yaml	PENDING - Document 3
17	Databricks connection	See Section 7	PENDING

5. Docker Compose Profiles

Profile	Services	Verified RAM	Start Command
registry	Local Docker registry	~11 MB	docker compose --profile registry up -d
storage	postgres, neo4j, qdrant, minio, redis	~904 MB	docker compose --profile storage up -d
messaging	kafka (KRaft), apicurio, redpanda-console	~618 MB (additional)	docker compose --profile storage --profile messaging up -d
streaming	flink-jm, flink-tm, debezium	~1.5 GB (additional)	Add --profile streaming
ai	ollama (load on demand)	~2.0 GB (on demand)	Add --profile ai
governance	SAM broker (deferred)	~1.0 GB	DEFERRED to Document 3
observability	otel, prometheus, grafana, loki, promtail, tempo, alertmanager	~1.9 GB (additional)	Add --profile observability

Resource Rule

Never run all profiles simultaneously. The storage + messaging combination (~1.52 GB) is sufficient for most development tasks and has been verified stable. Add observability only when debugging. The ai profile (Ollama) should be started on demand and stopped when not needed.

6. Python Virtual Environments

Venv	Activate	Key Packages	Purpose
.venv-langchain	source .venv-langchain/bin/activate	langgraph, langchain-groq, spacy>=3.8, neo4j, qdrant-client, fastapi	LangGraph agents, NLP pipeline, ingestion, graph, API
.venv-crewai	source .venv-crewai/bin/activate	crewai 0.80, langchain-groq	CrewAI specialist agent crews
.venv-dashboard	source .venv-dashboard/bin/activate	streamlit, plotly, pyvis, duckdb	Streamlit HITL demo application

The root pyproject.toml sets pythonpath=['.'] for pytest. Run all Python commands with PYTHONPATH=. from the project root.

7. Databricks Integration

7.1 Account and Token

- 1. Go to `community.cloud.databricks.com` - free account, no credit card required.
- 2. User Settings -> Access Tokens -> Generate New Token (90 days). Save as `DATABRICKS_TOKEN` in `.env`.
- 3. Note workspace URL from browser. Save as `DATABRICKS_HOST` in `.env`.

7.2 Data Flow: MinIO -> Iceberg -> Databricks

Step	Component	Operation
1. CDC	Debezium (streaming profile)	Captures row-level changes from PostgreSQL -> <code>arcas.cdc.*</code> Kafka topics
2. Lake write	Flink + Iceberg connector	Consumes CDC, writes Iceberg format files to MinIO <code>arcas-iceberg</code> bucket
3. Local query	DuckDB	Reads Iceberg tables from MinIO for local analytics
4. Cloud analytics	Databricks	Connects to MinIO via S3 API, reads Iceberg tables

Network Note

Databricks Community runs in the cloud. Your MinIO is behind NAT. Options: (a) ngrok http 9000 for a temporary tunnel; (b) Cloudflare R2 free tier (10 GB) as a permanent S3-compatible alternative.

8. Validation Checklist

#	Check	Command	Expected / Result
1	PostgreSQL healthy	<code>docker compose --profile storage ps postgres</code>	healthy - VERIFIED
2	14 tables created	<code>docker exec arcas-postgres psql -U arcas_app -d arcas -c '\dt'</code>	14 rows - VERIFIED
3	Audit protection	<code>docker exec arcas-postgres psql ... 'DELETE FROM audit_log'</code>	ERROR (expected) - VERIFIED
4	Neo4j healthy	<code>docker compose --profile storage ps neo4j</code>	healthy - VERIFIED
5	APOC installed	Neo4j browser: <code>CALL apoc.help('apoc')</code>	437 procedures - VERIFIED
6	Qdrant healthy	<code>docker compose --profile storage ps qdrant</code>	healthy - VERIFIED
7	MinIO healthy	<code>curl http://localhost:9000/minio/health/live</code>	HTTP 200 - VERIFIED
8	Redis healthy	<code>docker exec arcas-redis redis-cli ping</code>	PONG - VERIFIED
9	Kafka 30 topics	<code>docker exec arcas-kafka kafka-topics --list</code>	30 arcas.* topics - VERIFIED
10	MinIO 6 buckets	<code>mc ls arcas/</code>	6 buckets - VERIFIED
11	LangGraph importable	<code>python -c 'import langgraph'</code>	OK - VERIFIED
12	spaCy models	<code>python -c 'import spacy; spacy.load("es_core_news_lg")'</code>	OK - VERIFIED
13	Smoke test 7/7	<code>PYTHONPATH=. python scripts/maintenance/test_agent.py --synthetic --skip-llm</code>	7/7 PASS - VERIFIED
14	Redpanda Console	<code>http://localhost:8082</code>	1 broker online, 30 topics - VERIFIED
15	Total RAM (storage+messaging)	<code>docker stats --no-stream</code>	~1.52 GB - VERIFIED

Document 2 Complete

All 15 checks verified operational on 01 June 2026. Proceed to Document 3: Implementation.

End of Document 2 v3.0 - Environment Setup Guide - Verified Operational